# AERIS Expert Review Packet

## Anomaly

| | |
|---|---|
| Source / metric | tceq / so2 |
| Observed / expected | 19.3 / -0.310274 |
| Time | 2026-07-05 20:00 UTC |
| Location | 29.6240, -95.4743 |
| Severity | severe |
| Detectors | zscore, stl, isolation_forest |

## Stored evidence summary

| Source | Metric | Unit | Entities / points | Range; mean | Nearest to event |
|---|---|---|---|---|---|
| asos | humidity | percent | 9 / 869 | 36.74 to 100; mean 71.6752 | 46.79 at 2026-07-05 19:55Z; dt 5 min |
| asos | temperature | degC | 9 / 869 | 23.3333 to 36.6667; mean 29.6026 | 34 at 2026-07-05 19:55Z; dt 5 min |
| asos | wind_direction | degree | 9 / 1162 | 0 to 360; mean 149.69 | 190 at 2026-07-05 19:55Z; dt 5 min |
| asos | wind_speed | m/s | 9 / 1202 | 0 to 9.77444; mean 2.2756 | 2.05778 at 2026-07-05 19:55Z; dt 5 min |
| noaa_gfs | gh_500 | m | 11 / 132 | 5891.13 to 5930.39; mean 5910.21 | 5917.24 at 2026-07-05 18:00Z; dt 120 min |
| noaa_gfs | pbl_height | m | 11 / 132 | 35.9482 to 2074.66; mean 775.337 | 1770.74 at 2026-07-05 18:00Z; dt 120 min |
| noaa_gfs | precipitable_water | mm | 11 / 132 | 37.6598 to 50.8715; mean 44.2764 | 43.3105 at 2026-07-05 18:00Z; dt 120 min |
| noaa_gfs | surface_pressure | hPa | 11 / 132 | 1004.8 to 1015.62; mean 1011.82 | 1012.89 at 2026-07-05 18:00Z; dt 120 min |
| noaa_gfs | t_850 | degC | 11 / 132 | 17.3771 to 21.3086; mean 19.3479 | 17.5871 at 2026-07-05 18:00Z; dt 120 min |
| noaa_gfs | u_10m | m/s | 11 / 132 | -2.73237 to 3.62264; mean 0.6127 | 0.992425 at 2026-07-05 18:00Z; dt 120 min |
| noaa_gfs | v_10m | m/s | 11 / 132 | 0.329297 to 5.71323; mean 2.1821 | 1.9293 at 2026-07-05 18:00Z; dt 120 min |
| openaq | ozone | ppm | 16 / 1054 | -0.004 to 0.101; mean 0.017 | 0.027 at 2026-07-05 20:00Z; dt 0 min |
| openaq | pm10 | ug/m3 | 3 / 218 | 10 to 75; mean 19.8394 | 24 at 2026-07-05 20:00Z; dt 0 min |
| openaq | pm25 | ug/m3 | 80 / 3400 | 0 to 94.9; mean 6.7845 | 1.4 at 2026-07-05 20:00Z; dt 0 min |
| openweather | cloud_cover | percent | 5 / 359 | 0 to 100; mean 45.3733 | 100 at 2026-07-05 20:01Z; dt 1.9 min |
| openweather | humidity | percent | 5 / 359 | 50 to 95; mean 74.5738 | 57 at 2026-07-05 20:01Z; dt 1.9 min |
| openweather | precipitation | mm/h | 5 / 359 | 0 to 0.21; mean 0.001 | 0 at 2026-07-05 20:01Z; dt 1.9 min |
| openweather | pressure | hPa | 5 / 359 | 1011 to 1017; mean 1014.56 | 1014 at 2026-07-05 20:01Z; dt 1.9 min |
| openweather | temperature | degC | 5 / 359 | 25.15 to 36.43; mean 30.2805 | 34.99 at 2026-07-05 20:01Z; dt 1.9 min |
| openweather | wind_direction | degree | 5 / 359 | 0 to 340; mean 195.228 | 201 at 2026-07-05 20:01Z; dt 1.9 min |
| openweather | wind_speed | m/s | 5 / 359 | 0.45 to 6.09; mean 2.1281 | 1.29 at 2026-07-05 20:01Z; dt 1.9 min |
| purpleair | pm25 | ug/m3 | 71 / 5100 | 0 to 372.3; mean 8.382 | 3.1 at 2026-07-05 20:00Z; dt 0 min |

| Source | Metric | Unit | Entities / points | Range; mean | Nearest to event |
|---|---|---|---|---|---|
| sentinel5p | s5p_aer_ai_granule_available | count | 6 / 6 | 1 to 1; mean 1 | 1 at 2026-07-05 19:32Z; dt 27.4 min |
| sentinel5p | s5p_ch4_granule_available | count | 5 / 5 | 1 to 1; mean 1 | 1 at 2026-07-05 19:32Z; dt 27.4 min |
| sentinel5p | s5p_co_column | mol/m^2 | 6 / 6 | 0.0230724 to 0.0253077; mean 0.0245 | 0.0230724 at 2026-07-05 19:32Z; dt 27.4 min |
| sentinel5p | s5p_co_granule_available | count | 6 / 6 | 1 to 1; mean 1 | 1 at 2026-07-05 19:32Z; dt 27.4 min |
| sentinel5p | s5p_hcho_column | mol/m^2 | 5 / 5 | 0.000130424 to 0.000232421; mean 0.0002 | 0.000201778 at 2026-07-05 19:32Z; dt 27.4 min |
| sentinel5p | s5p_hcho_granule_available | count | 5 / 5 | 1 to 1; mean 1 | 1 at 2026-07-05 19:32Z; dt 27.4 min |
| sentinel5p | s5p_no2_column | mol/m^2 | 3 / 3 | 1.96909e-05 to 4.08591e-05; mean 0 | 1.96909e-05 at 2026-07-05 19:32Z; dt 27.4 min |
| sentinel5p | s5p_no2_granule_available | count | 3 / 3 | 1 to 1; mean 1 | 1 at 2026-07-05 19:32Z; dt 27.4 min |
| sentinel5p | s5p_o3_granule_available | count | 5 / 5 | 1 to 1; mean 1 | 1 at 2026-07-05 19:32Z; dt 27.4 min |
| sentinel5p | s5p_so2_column | mol/m^2 | 5 / 5 | -4.80456e-05 to 4.72775e-05; mean 0 | -4.80456e-05 at 2026-07-05 19:32Z; dt 27.4 min |
| sentinel5p | s5p_so2_granule_available | count | 5 / 5 | 1 to 1; mean 1 | 1 at 2026-07-05 19:32Z; dt 27.4 min |
| tceq | co | ppm | 3 / 188 | 0.1 to 0.8; mean 0.2144 | 0.2 at 2026-07-05 20:00Z; dt 0 min |
| tceq | no2 | ppb | 12 / 741 | -2.3 to 19.4; mean 3.502 | 2.7 at 2026-07-05 20:00Z; dt 0 min |
| tceq | so2 | ppb | 3 / 180 | -0.6 to 19.3; mean 0.2883 | 19.3 at 2026-07-05 20:00Z; dt 0 min |

## Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

| Source | Metric | Values |
|---|---|---|
| asos | humidity | 07-04 06Z 93.35, 12Z 67.57, 18Z 51.93, 07-05 00Z 75.63, 06Z 91.22, 12Z 69.81, 18Z 47.96, 07-06 00Z 73.47, 06Z 88.28, 12Z 70.19, 18Z 55.49, 07-07 00Z 77.29, 06Z 88.25 |
| asos | temperature | 07-04 06Z 25.32, 12Z 30.58, 18Z 33.91, 07-05 00Z 28.65, 06Z 25.68, 12Z 30.5, 18Z 34.07, 07-06 00Z 28.88, 06Z 26.06, 12Z 30.32, 18Z 32.62, 07-07 00Z 28.31, 06Z 26.09 |
| asos | wind_direction | 07-04 06Z 40.78, 12Z 189.3, 18Z 142, 07-05 00Z 164.4, 06Z 79.15, 12Z 201.8, 18Z 203.9, 07-06 00Z 153.8, 06Z 119.4, 12Z 180.9, 18Z 166.7, 07-07 00Z 137.1, 06Z 96.25 |
| asos | wind_speed | 07-04 06Z 0.4917, 12Z 2.202, 18Z 3.224, 07-05 00Z 3.081, 06Z 0.7639, 12Z 2.378, 18Z 2.71, 07-06 00Z 3.066, 06Z 1.243, 12Z 2.058, 18Z 3.818, 07-07 00Z 2.083, 06Z 1.044 |
| noaa_gfs | gh_500 | 07-04 12Z 5919, 18Z 5930, 07-05 00Z 5918, 06Z 5923, 12Z 5907, 18Z 5917, 07-06 00Z 5905, 06Z 5907, 12Z 5892, 18Z 5902, 07-07 00Z 5898, 06Z 5905 |
| noaa_gfs | pbl_height | 07-04 12Z 117.3, 18Z 1856, 07-05 00Z 834.9, 06Z 289.5, 12Z 161.5, 18Z 1764, 07-06 00Z 831.9, 06Z 261.3, 12Z 169.2, 18Z 1793, 07-07 00Z 993.8, 06Z 231.7 |
| noaa_gfs | precipitable_water | 07-04 12Z 42.15, 18Z 45.88, 07-05 00Z 49.22, 06Z 43.89, 12Z 42.35, 18Z 42.98, 07-06 00Z 43.05, 06Z 44.67, 12Z 41.87, 18Z 43.04, 07-07 00Z 46.6, 06Z 45.61 |
| noaa_gfs | surface_pressure | 07-04 12Z 1013, 18Z 1013, 07-05 00Z 1011, 06Z 1012, 12Z 1013, 18Z 1012, 07-06 00Z 1009, 06Z 1011, 12Z 1012, 18Z 1012, 07-07 00Z 1011, 06Z 1013 |
| noaa_gfs | t_850 | 07-04 12Z 19.55, 18Z 18.12, 07-05 00Z 19.78, 06Z 19.75, 12Z 19.4, 18Z 18.01, 07-06 00Z 20.22, 06Z 20.13, 12Z 19.66, 18Z 17.99, 07-07 00Z 19.79, 06Z 19.78 |
| noaa_gfs | u_10m | 07-04 12Z 0.9164, 18Z 0.8209, 07-05 00Z -1.26, 06Z 1.476, 12Z 1.55, 18Z 0.9106, 07-06 00Z 0.1863, 06Z 1.334, 12Z 1.072, 18Z 0.5889, 07-07 00Z -0.636, 06Z 0.3935 |

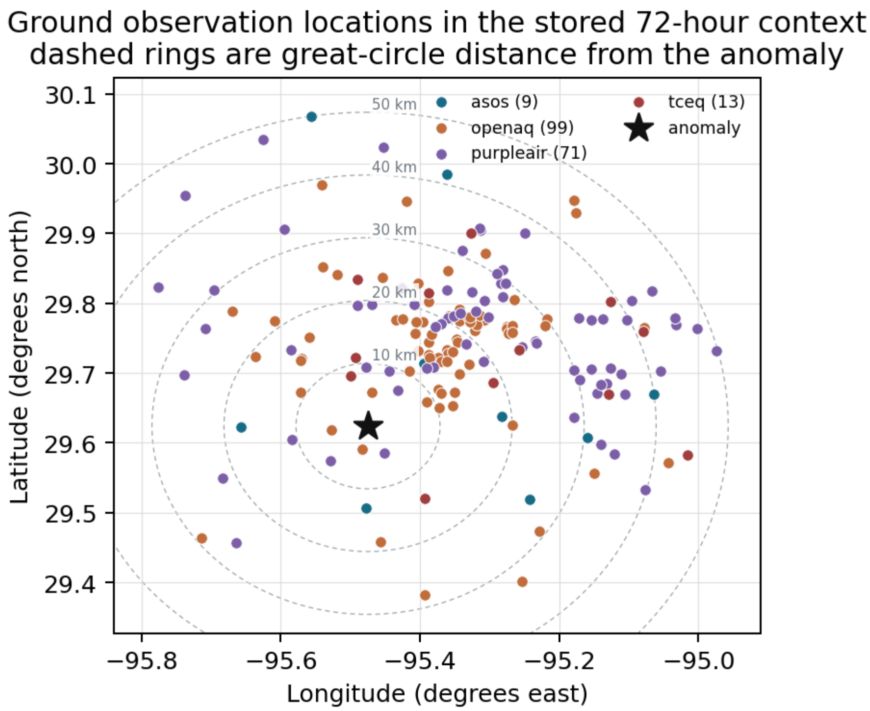
| Source | Metric | Values |
|---|---|---|
| noaa_gfs | v_10m | 07-04 12Z 1.087, 18Z 1.721, 07-05 00Z 3.911, 06Z 2.442, 12Z 1.029, 18Z 1.484, 07-06 00Z 3.259, 06Z 2.538, 12Z 1.305, 18Z 1.62, 07-07 00Z 3.61, 06Z 2.18 |
| openaq | ozone | 07-04 06Z 0.006033, 12Z 0.01404, 18Z 0.03177, 07-05 00Z 0.01623, 06Z 0.006943, 12Z 0.01344, 18Z 0.02877, 07-06 00Z 0.02044, 06Z 0.00661, 12Z 0.0121, 18Z 0.03261, 07-07 00Z 0.01312, 06Z 0.006162 |
| openaq | pm10 | 07-04 06Z 18.5, 12Z 16.83, 18Z 17.28, 07-05 00Z 22.39, 06Z 24.78, 12Z 18.72, 18Z 16.59, 07-06 00Z 15.89, 06Z 14.39, 12Z 30.22, 18Z 27.5, 07-07 00Z 15.89, 06Z 16.78 |
| openaq | pm25 | 07-04 06Z 5.451, 12Z 5.096, 18Z 5.347, 07-05 00Z 11.49, 06Z 15.52, 12Z 6.637, 18Z 4.844, 07-06 00Z 5.394, 06Z 6.698, 12Z 6.37, 18Z 5.38, 07-07 00Z 4.58, 06Z 4.921 |
| openweather | cloud_cover | 07-04 06Z 7, 12Z 2.448, 18Z 38.74, 07-05 00Z 89.97, 06Z 89.03, 12Z 28.23, 18Z 92.41, 07-06 00Z 88.13, 06Z 41.86, 12Z 24.47, 18Z 24.9, 07-07 00Z 14.61, 06Z 11.11 |
| openweather | humidity | 07-04 06Z 90.75, 12Z 77.9, 18Z 58.42, 07-05 00Z 74.5, 06Z 87.73, 12Z 78.83, 18Z 57.69, 07-06 00Z 72.74, 06Z 86.38, 12Z 78.2, 18Z 59.53, 07-07 00Z 75.58, 06Z 83 |
| openweather | precipitation | 07-04 06Z 0, 12Z 0, 18Z 0, 07-05 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-06 00Z 0.006774, 06Z 0, 12Z 0, 18Z 0.005333, 07-07 00Z 0, 06Z 0 |
| openweather | pressure | 07-04 06Z 1016, 12Z 1017, 18Z 1014, 07-05 00Z 1014, 06Z 1015, 12Z 1016, 18Z 1013, 07-06 00Z 1013, 06Z 1014, 12Z 1016, 18Z 1014, 07-07 00Z 1014, 06Z 1016 |
| openweather | temperature | 07-04 06Z 26.1, 12Z 29.88, 18Z 35.03, 07-05 00Z 30.05, 06Z 26.68, 12Z 29.88, 18Z 34.99, 07-06 00Z 30.2, 06Z 26.53, 12Z 29.63, 18Z 34.39, 07-07 00Z 29.5, 06Z 26.93 |
| openweather | wind_direction | 07-04 06Z 197.4, 12Z 223.3, 18Z 171.7, 07-05 00Z 169.9, 06Z 194.7, 12Z 216.4, 18Z 196.2, 07-06 00Z 175.9, 06Z 231.1, 12Z 233.5, 18Z 169.2, 07-07 00Z 178.6, 06Z 160.6 |
| openweather | wind_speed | 07-04 06Z 1.59, 12Z 1.607, 18Z 2.431, 07-05 00Z 2.49, 06Z 1.986, 12Z 1.85, 18Z 2.166, 07-06 00Z 2.95, 06Z 2.097, 12Z 1.838, 18Z 2.215, 07-07 00Z 2.186, 06Z 1.776 |
| purpleair | pm25 | 07-04 06Z 7.367, 12Z 6.332, 18Z 5.958, 07-05 00Z 15.16, 06Z 15.45, 12Z 6.761, 18Z 5.493, 07-06 00Z 8.579, 06Z 9.001, 12Z 8.288, 18Z 6.302, 07-07 00Z 6.214, 06Z 7.07 |
| sentinel5p | s5p_aer_ai_granule_available | 07-04 18Z 1, 07-05 18Z 1, 07-06 18Z 1 |
| sentinel5p | s5p_ch4_granule_available | 07-04 18Z 1, 07-05 18Z 1, 07-06 18Z 1 |
| sentinel5p | s5p_co_column | 07-04 18Z 0.02512, 07-05 18Z 0.02308, 07-06 18Z 0.02528 |
| sentinel5p | s5p_co_granule_available | 07-04 18Z 1, 07-05 18Z 1, 07-06 18Z 1 |
| sentinel5p | s5p_hcho_column | 07-04 18Z 0.0001321, 07-05 18Z 0.0002018, 07-06 18Z 0.0002254 |
| sentinel5p | s5p_hcho_granule_available | 07-04 18Z 1, 07-05 18Z 1, 07-06 18Z 1 |
| sentinel5p | s5p_no2_column | 07-04 18Z 2.303e-05, 07-05 18Z 1.969e-05, 07-06 18Z 4.086e-05 |
| sentinel5p | s5p_no2_granule_available | 07-04 18Z 1, 07-05 18Z 1, 07-06 18Z 1 |
| sentinel5p | s5p_o3_granule_available | 07-04 18Z 1, 07-05 18Z 1, 07-06 18Z 1 |
| sentinel5p | s5p_so2_column | 07-04 18Z 2.773e-06, 07-05 18Z -4.805e-05, 07-06 18Z 4.559e-05 |
| sentinel5p | s5p_so2_granule_available | 07-04 18Z 1, 07-05 18Z 1, 07-06 18Z 1 |
| tceq | co | 07-04 06Z 0.225, 12Z 0.2167, 18Z 0.2222, 07-05 00Z 0.24, 06Z 0.1944, 12Z 0.2, 18Z 0.2071, 07-06 00Z 0.21, 06Z 0.25, 12Z 0.2353, 18Z 0.1882, 07-07 00Z 0.23, 06Z 0.1556 |
| tceq | no2 | 07-04 06Z 4.562, 12Z 2.953, 18Z 2.513, 07-05 00Z 1.951, 06Z 3.896, 12Z 2.649, 18Z 2.067, 07-06 00Z 2.523, 06Z 4.436, 12Z 5.197, 18Z 4.293, 07-07 00Z 3.786, 06Z 4.909 |
| tceq | so2 | 07-04 06Z -0.2167, 12Z 0.02778, 18Z 0.5444, 07-05 00Z -0.1333, 06Z -0.2437, 12Z 0.1706, 18Z 3.087, 07-06 00Z 0.11, 06Z -0.2, 12Z 0.1353, 18Z 0.075, 07-07 00Z -0.23, 06Z -0.3125 |

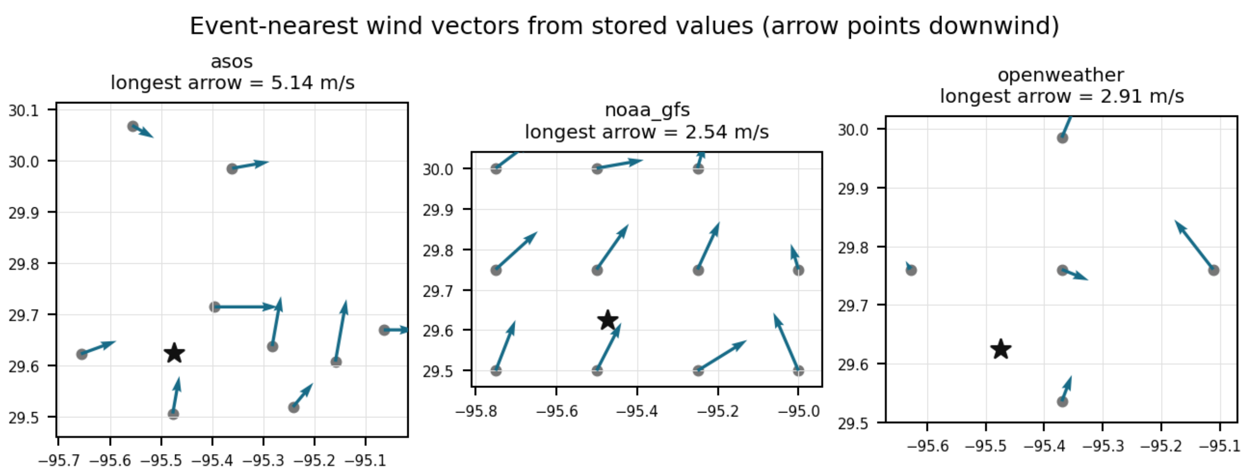
## Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

| Source | Metric | Values |
|---|---|---|
| asos | humidity | MCJ (12.608 km) 69.97, AXH (13.11 km) 71.68, SGR (17.608 km) 75.52, HOU (18.613 km) 70.42, LVJ (25.353 km) 71.98, EFD (30.569 km) 71.81, T41 (39.963 km) 71.65, IAH (41.55 km) 68.86, +1 more |
| asos | temperature | MCJ (12.608 km) 29.92, AXH (13.11 km) 28.99, SGR (17.608 km) 29.33, HOU (18.613 km) 30.3, LVJ (25.353 km) 29.55, EFD (30.569 km) 29.87, T41 (39.963 km) 29.78, IAH (41.55 km) 30.15, +1 more |
| asos | wind_direction | MCJ (12.608 km) 142.8, AXH (13.11 km) 88.94, SGR (17.608 km) 166.3, HOU (18.613 km) 176.6, LVJ (25.353 km) 123.7, EFD (30.569 km) 150.3, T41 (39.963 km) 189.1, IAH (41.55 km) 166.8, +1 more |
| asos | wind_speed | MCJ (12.608 km) 3.006, AXH (13.11 km) 0.9851, SGR (17.608 km) 2.522, HOU (18.613 km) 2.819, LVJ (25.353 km) 1.747, EFD (30.569 km) 2.8, T41 (39.963 km) 2.889, IAH (41.55 km) 2.358, +1 more |
| noaa_gfs | gh_500 | gfs:29.50,-95.50 (14.007 km) 5911, gfs:29.75,-95.50 (14.23 km) 5910, gfs:29.50,-95.25 (25.708 km) 5910, gfs:29.75,-95.25 (25.808 km) 5910, gfs:29.50,-95.75 (30.014 km) 5911, gfs:29.75,-95.75 (30.09 km) 5910, gfs:30.00,-95.50 (41.885 km) 5910, gfs:30.00,-95.25 (47.082 km) 5910, +3 more |
| noaa_gfs | pbl_height | gfs:29.50,-95.50 (14.007 km) 648.6, gfs:29.75,-95.50 (14.23 km) 970.7, gfs:29.50,-95.25 (25.708 km) 632, gfs:29.75,-95.25 (25.808 km) 935.8, gfs:29.50,-95.75 (30.014 km) 592.1, gfs:29.75,-95.75 (30.09 km) 660.1, gfs:30.00,-95.50 (41.885 km) 1017, gfs:30.00,-95.25 (47.082 km) 1028, +3 more |
| noaa_gfs | precipitable_water | gfs:29.50,-95.50 (14.007 km) 43.68, gfs:29.75,-95.50 (14.23 km) 44.37, gfs:29.50,-95.25 (25.708 km) 44.4, gfs:29.75,-95.25 (25.808 km) 45.07, gfs:29.50,-95.75 (30.014 km) 43.01, gfs:29.75,-95.75 (30.09 km) 42.8, gfs:30.00,-95.50 (41.885 km) 45.13, gfs:30.00,-95.25 (47.082 km) 45.99, +3 more |
| noaa_gfs | surface_pressure | gfs:29.50,-95.50 (14.007 km) 1012, gfs:29.75,-95.50 (14.23 km) 1012, gfs:29.50,-95.25 (25.708 km) 1013, gfs:29.75,-95.25 (25.808 km) 1013, gfs:29.50,-95.75 (30.014 km) 1012, gfs:29.75,-95.75 (30.09 km) 1010, gfs:30.00,-95.50 (41.885 km) 1010, gfs:30.00,-95.25 (47.082 km) 1011, +3 more |
| noaa_gfs | t_850 | gfs:29.50,-95.50 (14.007 km) 19.39, gfs:29.75,-95.50 (14.23 km) 19.33, gfs:29.50,-95.25 (25.708 km) 19.37, gfs:29.75,-95.25 (25.808 km) 19.24, gfs:29.50,-95.75 (30.014 km) 19.28, gfs:29.75,-95.75 (30.09 km) 19.38, gfs:30.00,-95.50 (41.885 km) 19.35, gfs:30.00,-95.25 (47.082 km) 19.22, +3 more |
| noaa_gfs | u_10m | gfs:29.50,-95.50 (14.007 km) 0.4203, gfs:29.75,-95.50 (14.23 km) 0.717, gfs:29.50,-95.25 (25.708 km) 0.412, gfs:29.75,-95.25 (25.808 km) 0.652, gfs:29.50,-95.75 (30.014 km) 0.2536, gfs:29.75,-95.75 (30.09 km) 0.882, gfs:30.00,-95.50 (41.885 km) 1.379, gfs:30.00,-95.25 (47.082 km) 0.09029, +3 more |
| noaa_gfs | v_10m | gfs:29.50,-95.50 (14.007 km) 2.242, gfs:29.75,-95.50 (14.23 km) 2.393, gfs:29.50,-95.25 (25.708 km) 2.18, gfs:29.75,-95.25 (25.808 km) 2.478, gfs:29.50,-95.75 (30.014 km) 2.071, gfs:29.75,-95.75 (30.09 km) 2.337, gfs:30.00,-95.50 (41.885 km) 1.627, gfs:30.00,-95.25 (47.082 km) 2.063, +3 more |
| openaq | ozone | 312 (0.017 km) 0.01684, 311 (8.339 km) 0.01688, 3055 (13.984 km) 0.01324, 1319333 (18.217 km) 0.02259, 325 (18.693 km) 0.004941, 287 (19.112 km) 0.01784, 320 (20.024 km) 0.01289, 319 (23.419 km) 0.01733, +8 more |
| openaq | pm10 | 15852419 (22.104 km) 21.96, 13380082 (24.254 km) 21.78, 271 (33.811 km) 15.72 |
| openaq | pm25 | 13955814 (3.744 km) 4.75, 15399615 (5.167 km) 4.452, 8967119 (5.456 km) 8.297, 14140752 (8.986 km) 7.452, 8967021 (10.262 km) 8.825, 8967001 (10.484 km) 5.768, 13955816 (10.805 km) 7.138, 15539683 (11.385 km) 8.113, +72 more |
| openweather | cloud_cover | grid:south (14.084 km) 46.51, grid:center (18.223 km) 45.66, grid:west (21.256 km) 46.99, grid:east (38.226 km) 38.75, grid:north (41.392 km) 48.96 |
| openweather | humidity | grid:south (14.084 km) 73.85, grid:center (18.223 km) 73.32, grid:west (21.256 km) 73.74, grid:east (38.226 km) 78.93, grid:north (41.392 km) 73.01 |
| openweather | precipitation | grid:south (14.084 km) 0, grid:center (18.223 km) 0.002254, grid:west (21.256 km) 0, grid:east (38.226 km) 0, grid:north (41.392 km) 0.002917 |
| openweather | pressure | grid:south (14.084 km) 1015, grid:center (18.223 km) 1014, grid:west (21.256 km) 1015, grid:east (38.226 km) 1015, grid:north (41.392 km) 1014 |
| openweather | temperature | grid:south (14.084 km) 30.1, grid:center (18.223 km) 30.78, grid:west (21.256 km) 30.28, grid:east (38.226 km) 29.92, grid:north (41.392 km) 30.33 |
| openweather | wind_direction | grid:south (14.084 km) 201, grid:center (18.223 km) 215.1, grid:west (21.256 km) 189.1, grid:east (38.226 km) 195.1, grid:north (41.392 km) 176.1 |

| Source | Metric | Values |
| --- | --- | --- |
| openweather | wind_speed | grid:south (14.084 km) 2.113, grid:center (18.223 km) 1.915, grid:west (21.256 km) 1.786, grid:east (38.226 km) 2.888, grid:north (41.392 km) 1.935 |
| purpleair | pm25 | 161003 (4.856 km) 5.878, 166451 (7.01 km) 8.99, 161015 (7.666 km) 9.126, 46237 (9.249 km) 0.1764, 186315 (9.424 km) 8.832, 276512 (10.819 km) 7.621, 293735 (12.303 km) 9.464, 293725 (13.024 km) 9.483, +63 more |
| tceq | co | 482011052 (22.762 km) 0.3141, 482011035 (24.236 km) 0.1806, 482011039 (33.805 km) 0.1452 |
| tceq | no2 | 482010055 (8.334 km) 2.623, 482011066 (11.0 km) 4.087, 480391004 (13.97 km) 1.465, 482010416 (18.688 km) 3.773, 482011052 (22.762 km) 6.865, 482010047 (23.42 km) 3.732, 482011035 (24.236 km) 4.694, 482011039 (33.805 km) 2.183, +4 more |
| tceq | so2 | 482010051 (0.0 km) 0.8429, 482011035 (24.236 km) 0.006557, 482011039 (33.805 km) -0.02857 |



## Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

### Claim 1 of 35

Sentinel-5P satellite measurements at 19:32 UTC did not capture elevated column SO2, indicating a localized ground-level plume below the satellite spatial resolution or detection limit.

| **Mark exactly one:** | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 2 of 35

The anomaly was likely caused by a combination of factors, including weather patterns and local emissions.

| **Mark exactly one:** | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 3 of 35

The air quality anomaly in Houston, Texas is caused by high levels of SO2 particles.

| **Mark exactly one:** | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 4 of 35

An unpermitted industrial upset, degassing, or flaring release from a petrochemical or refining facility upwind of TCEQ station 482010051 caused the isolated sulfur dioxide surge.

| **Mark exactly one:** | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 5 of 35

On July 5, 2026, at 20:00:00 UTC, sulfur dioxide concentration recorded by TCEQ monitor 482010051 at coordinates (29.624, -95.474) reached a severe spike of 19.3 ppb compared to an expected baseline of -0.31 ppb.

| **Mark exactly one:** | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 6 of 35

The TCEQ monitor at the anomaly location recorded a severe SO2 spike of 19.3 ppb at 2026-07-05T20:00:00+00:00, while the other nearby TCEQ SO2 monitors had much lower mean SO2 levels during the context window, indicating a strongly localized event rather than a metro-wide SO2 episode.

| **Mark exactly one:** | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 7 of 35

The tceq SO2 monitor at 29.624, -95.474 recorded 19.3 ppb at 2026-07-05T20:00:00+00:00.

| **Mark exactly one:** | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 8 of 35

The anomaly occurred on July 5th at around 20:00 UTC.

| Mark exactly one: | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 9 of 35

Co-pollutant and satellite indicators did not show a corresponding regional combustion or aerosol event at the anomaly time, because NO2, CO, PM2.5, PM10, Sentinel-5P CO, Sentinel-5P NO2, and Sentinel-5P SO2 column data were not elevated in a way matching the surface SO2 spike.

| Mark exactly one: | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 10 of 35

A broad regional combustion or aerosol episode is contradicted because co-pollutants and satellite fields were not elevated near the anomaly time, including low TCEQ NO2 near 2.7 ppb, modest TCEQ CO near 0.2 ppm, low nearby PM2.5 of 1.4 ug/m3 in openaq and 3.1 ug/m3 in purpleair, and no positive Sentinel-5P SO2 column enhancement over the area.

| Mark exactly one: | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 11 of 35

The temperature was unusually high on July 5th, potentially contributing to the air quality anomaly.

| **Mark exactly one:** | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 12 of 35

The tceq SO2 anomaly at 2026-07-05T20:00:00+00:00 had an expected value of -0.3102739726027397 ppb and a z-score of 19.687653620618416, and the anomaly severity was classified as severe.

| **Mark exactly one:** | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 13 of 35

Meteorological transport and limited near-source dilution likely contributed to the spike because the anomaly occurred during hot afternoon conditions with modest southerly to south-southwesterly winds of about 1 to 2 m/s near the site, which can carry a narrow plume from industrial areas into a monitor without requiring region-wide accumulation.

| **Mark exactly one:** | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 14 of 35

The presence of high levels of PM2.5 particles in the atmosphere on July 5th, as measured by PurpleAir sensors, may have contributed to the air quality anomaly.

| Mark exactly one: | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 15 of 35

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

| Mark exactly one: | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 16 of 35

ASOS surface weather stations near the anomaly recorded light south-southwesterly winds around 190 to 204 degrees at approximately 2 m/s around 20:00 UTC on July 5, 2026.

**Calm-wind flag: wind direction unstable under calm conditions. openweather: event wind 1.29 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave -0.300086308282022 m/s; n=359 wind readings). Cutoff floor: confirmed 2026-07-24.**

| Mark exactly one: | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 17 of 35

During the 6-hour window from 18:00 UTC on July 5, 2026, to 00:00 UTC on July 6, 2026, the local TCEQ 6-hour mean $SO_2$ concentration rose to 3.087 ppb, driven entirely by the localized spike at station 482010051 while regional monitors over 24 km away reported mean concentrations near 0.0 ppb.

| **Mark exactly one:** | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 18 of 35

Southerly to south-southwesterly surface winds transported the localized industrial sulfur dioxide plume directly across the monitoring location during the afternoon.

| **Mark exactly one:** | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 19 of 35

The air quality anomaly in Houston, Texas on July 5th was characterized by elevated levels of $PM_{2.5}$.

| **Mark exactly one:** | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 20 of 35

A short-duration industrial sulfur emission plume from the Houston Ship Channel or nearby refinery-petrochemical sources is the most plausible physical cause of the 19.3 ppb $SO_2$ spike because the anomaly was extremely large at one TCEQ $SO_2$ monitor while other regional pollutants remained low to moderate and nearby $SO_2$ background at the other TCEQ monitors was near zero.

| **Mark exactly one:** | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 21 of 35

The PM2.5 levels are elevated at 3.1 ug/m3, which is higher than the mean of 8.382 ug/m3.

| **Mark exactly one:** | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 22 of 35

A strong wind from the west or northwest direction on July 5th may have dispersed pollutants, leading to the air quality anomaly.

**Calm-wind flag: wind direction unstable under calm conditions. openweather: event wind 1.29 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave -0.300086308282022 m/s; n=359 wind readings). Cutoff floor: confirmed 2026-07-24.**

| **Mark exactly one:** | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 23 of 35

Surface and boundary-layer meteorology near Houston at the anomaly time featured warm afternoon conditions, light to modest southerly to south-southwesterly flow, no precipitation, and a deep planetary boundary layer, which is consistent with advection of a nearby plume rather than stagnant regional accumulation.

| **Mark exactly one:** | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 24 of 35

The air quality anomaly is related to PM2.5 levels.

| **Mark exactly one:** | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 25 of 35

A regional combustion or smoke episode is an unlikely primary cause of the SO2 anomaly because PM2.5, PM10, CO, NO2, satellite CO, and satellite NO2 were not elevated at the time, indicating the signal was dominated by sulfur-rich local emissions rather than widespread urban or fire-related pollution.

| **Mark exactly one:** | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 26 of 35

The nearest other tceq $SO_2$ monitors within 50 km had period-mean $SO_2$ values of 0.006557 ppb at 24.236 km and -0.02857 ppb at 33.805 km, while the anomalous monitor had a period-mean $SO_2$ value of 0.8429 ppb.

| **Mark exactly one:** | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 27 of 35

Vigorous daytime boundary layer convection with boundary layer heights near 1,770 meters entrained elevated industrial stack plumes down to ground-level sensors.

| **Mark exactly one:** | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 28 of 35

The air quality anomaly in Houston, Texas is caused by high levels of $NO_2$ particles.

| **Mark exactly one:** | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 29 of 35

$NO_2$ levels were also elevated during the same period.

| **Mark exactly one:** | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 30 of 35

A local short-duration sulfur emission plume is strongly supported because the TCEQ SO2 monitor at the anomaly site measured 19.3 ppb at 2026-07-05T20:00:00+00:00 while the other two TCEQ SO2 monitors in the 50 km domain had near-zero mean SO2 over the context window, indicating strong spatial localization.

**Calm-wind flag: wind direction unstable under calm conditions. openweather: event wind 1.29 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave -0.300086308282022 m/s; n=359 wind readings). Cutoff floor: confirmed 2026-07-24.**

| **Mark exactly one:** | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 31 of 35

The sulfur dioxide spike reaching 19.3 ppb at monitor 482010051 aligns with an upwind point-source emission driven by south-southwesterly surface winds recorded by ASOS and OpenWeather.

**Calm-wind flag: wind direction unstable under calm conditions. openweather: event wind 1.29 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave -0.300086308282022 m/s; n=359 wind readings). Cutoff floor: confirmed 2026-07-24.**

| **Mark exactly one:** | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 32 of 35

NOAA GFS model data indicated a deep daytime planetary boundary layer height exceeding 1700 meters near the anomaly location at 18:00 UTC on July 5, 2026.

| **Mark exactly one:** | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 33 of 35

TCEQ monitor 482010051 recorded a sulfur dioxide concentration of 19.3 ppb at 20:00 UTC on July 5, 2026, while other regional TCEQ monitors observed baseline concentrations near 0 ppb.

| Mark exactly one: | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 34 of 35

Vertical mixing within the lower atmosphere favoured ground-level accumulation of emissions, supported by GFS data showing a planetary boundary layer height near 1,764 meters around the time of the event.

| Mark exactly one: | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

### Claim 35 of 35

Transport from sources south to south-southwest of the anomaly site is supported because near the anomaly time local winds were about 190 to 201 degrees at roughly 1.3 to 2.1 m/s and the afternoon boundary layer was deep at about 1771 m, conditions consistent with advection of a nearby plume rather than nocturnal trapping.

| Mark exactly one: | [ ] V | [ ] I | [ ] U |
|---|---|---|---|

**Note:**

## Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.